

CapSense: broadcast GPS position every 60 s

App-only setup. Starting point: node already on the **LONG_MODERATE** preset, role **CLIENT**.

1 Find the Position screen

This app version has **no "Radio configuration"** entry. The Settings screen opens on a group called **Configuration**, and the position settings are nested inside **Device configuration** — not in *Module configuration* (MQTT, Serial, Telemetry...).

Configuration	Settings screen
User	
LoRa	region · preset · TX
Channels	step 3
Security	
Device configuration	open this → steps 2 and 4
Module configuration	not here
Administration	

Settings > **Device configuration** > Device · **Position** · Power · Network · Display · Bluetooth

So: **Position** and **Power** are inside **Device configuration**; **Channels** and **LoRa** are on the Settings screen itself.

2 Position — the heartbeat

Settings > Device configuration > **Position**

CARD	SETTING	VALUE	WHY
Position Packet	Broadcast Interval	1 minute	The heartbeat. It is a dropdown, not a number field — the options are Unset, 1 minute , 90 seconds, 5 minutes, 15 minutes, 1 hour... Default (Unset) is 15 minutes.
Position Packet	Smart Position	OFF	ON by default. It adds extra transmissions whenever you move ≥ 100 m, on top of the 1-minute beacon. Off = exactly one packet per minute.
Device GPS	Fixed Position	OFF	Would ignore the real GNSS.
Device GPS	GPS Mode	Enabled	Not <i>Disabled</i> , not <i>Not present</i> .
Device GPS	Update Interval	40 seconds	Also a dropdown, and it has no 30 s option : Unset, 8 s, 20 s, 40 s , 1 minute, 80 s, 2 minutes... Pick 40 s so every beacon carries a fresh fix. The default (2 minutes) would repeat stale coordinates.
Position Flags	Position Flags	Minimum	Each flag adds bytes to 1,440 packets/day.
Advanced Device GPS	RX / TX / EN GPIO	Don't touch	Set by the stock firmware.

Tap **Save**. The node reboots and BLE drops for a few seconds — normal. Nothing applies until you save.

3 Channel

Settings > **Channels** > primary channel (index 0)

SETTING	VALUE	WHY
Position enabled	ON	Off sets the channel's position precision to 0 — the beacon carries no coordinates at all.
Precise location	ON	Appears once <i>Position enabled</i> is on. Turning it off replaces exact coordinates with a privacy radius (a slider showing $\pm$ distance).

Save. Do **not** change the channel name or PSK — the node would drop off your mesh.

4 Power

Settings > Device configuration > **Power**

SETTING	VALUE	WHY
Power Saving	OFF	Sleep cycles cut the GNSS between beacons and make the heartbeat irregular.

Save. That is the last configuration screen.

5 Verify

- Outdoors, wait for the first fix — **1 to 5 minutes** on a cold start. The very first beacon after boot is skipped until a fresh fix exists.
- Nodes** tab > your CapSense: coordinates and a recent timestamp.
- Watch **last heard** for 3–4 minutes: it must reset **every ~60 s**.
- Real check:** connect the app to a **different node** and confirm the CapSense shows up there refreshing every minute. That proves the LoRa path, not just Bluetooth.

Summary

SCREEN	SETTING	VALUE
Device config > Position	GPS Mode	Enabled
Device config > Position	Broadcast Interval	1 minute
Device config > Position	Smart Position	OFF
Device config > Position	Fixed Position	OFF
Device config > Position	Update Interval	40 seconds
Channels > 0	Position enabled · Precise location	ON · ON
Device config > Power	Power Saving	OFF
LoRa	Region · Modem Preset · TX	Yours · Long Range - Moderate · ON
Device config > Device	Role	Client

Why it may still not beacon every 60 s

Two firmware rules override your setting. Neither shows an error in the app — the beacon just doesn't go out.

RULE	EFFECT
Channel utilisation gate	Position packets are dropped while channel utilisation is $\geq 25\%$ for the Client role (Tracker roles get 40 %). A busy mesh silently stops your heartbeat.
Node-count scaling	Above 40 online nodes the firmware stretches the interval automatically. On LONG_MODERATE the factor is $1 + 0.16$ per extra node: 50 nodes $\rightarrow \sim 2.6\times$ (≈ 158 s), 60 nodes $\rightarrow \sim 4.3\times$ (≈ 257 s) . You cannot hold 60 s on a large mesh.

SYMPTOM	FIX
Node visible, no position	No GPS fix. Go outdoors, wait 5 min. Confirm GPS Mode = Enabled.
Beacons every 15 min	Broadcast Interval is still Unset (= 15 min), or Save didn't apply. Reconnect and re-check the dropdown.
Extra beacons while moving	Smart Position still ON.
Same coordinates repeated	Update Interval is above 1 minute. Set it to 40 seconds.
Other nodes don't see it	Channel <i>Position enabled</i> is off, or mismatched region / preset / channel PSK, or TX disabled.

COST ON LONG_MODERATE

0.34 kbps: each position packet holds the channel for roughly 1.5-2 s, so one node alone sits near 2-3 % utilisation continuously, multiplied by every rebroadcast and hop — which feeds straight into the 25 % gate above. Battery with the GNSS always on lasts hours, not days. Use 60 s only on the node you are actively tracking; 120-300 s otherwise.

REVERT

Broadcast Interval back to *Unset* (15 minutes), Smart Position back ON. Menu paths, field names and dropdown values verified against the Meshtastic Android app source (Sept 2026); airtime figures are estimates for a typical position packet.